Marathwada Shikshan Prasarak Mandal's

Deogiri Institute of Engineering and Management Studies, Aurangabad

Project Report

On

Sign Language Recognition (SLR)

Submitted By

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(2021 - 2022)

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In partial fulfilment of

Bachelor of Technology

(Computer Science & Engineering)

Guided By

Mrs. Smita Ponde

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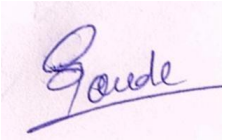
(2021- 2022)

CERTIFICATE

This is to certify that, the Project entitled "**Sign Language Recognition (SLR)**" submitted by **Richa Vijay Patil, Riteeka Amol Purnekar, Nilesh Sharad Mahajan** is a bonafide work completed under my supervision and guidance in partial fulfilment for award of Bachelor of Technology (Computer Science and Engineering) Degree of Dr. Babasaheb Ambedkar Technological University, Lonere.

Place: Aurangabad

Date: 15/01/2022

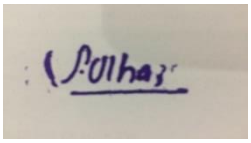


Mrs. Smita Ponde

Guide

Mr. S.B. Kalyankar

Head



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DECLARATION

This is to certify that, the partial project report entitled (TNRB 12), "**Sign Language Recognition (SLR)**" Submitted by **Richa Vijay Patil, Riteeka Amol Purnekar, Nilesh Sharad Mahajan** is a bonafide work completed under my supervision and guidance in partial fulfilment for award of Bachelor degree in Computer Science and Engineering of Deogiri Institute of Engineering and Management Studies, Aurangabad under Dr. Babasaheb Ambedkar Technological University, Lonere.

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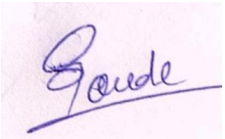

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ABSTRACT

Sign language is the preferred method of communication among the deaf and the hearing-impaired people all over the world. Recognition of sign language can have varying degree of success when used in a computer vision or any other methods. Sign language is said to have a structured set of gestures in which each gesture is having a specific meaning. To make sure that the deaf people receive proper education in sign language as well as get employed we are making a Sign Recognition System to help them. The technologies and tools used in our project are Python as a Programming language, TensorFlow and Keras as Deep Learning libraries and OpenCV as a main component.

Hand gesture is one of the methods used in sign language for non-verbal communication. It is most commonly used by deaf & dumb people who have hearing or speech problems to communicate among themselves or with normal people. Various sign language systems have been developed by many makers around the world but they are neither flexible nor cost-effective for the end users. Hence in this paper introduced software which presents a system prototype that is able to automatically recognize sign language to help deaf and dumb people to communicate more effectively with each other or normal people.

Pattern recognition and Gesture recognition are the developing fields of research. Being a significant part in nonverbal communication hand gestures are playing key role in our daily life. Hand Gesture recognition system provides us an innovative, natural, user friendly way of communication with the computer which is more familiar to the human beings. By considering in mind the similarities of human hand shape with four fingers and one thumb, the software aims to present a real time system for recognition of hand gesture.

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1. INTRODUCTION

1.1 Introduction

This project aims at identifying alphabets in American Sign Language from the corresponding gestures. Gesture recognition and sign language recognition has been a well-researched topic for Indian Sign Language (ASL), but few research works have been published regarding American Sign Language (ASL). But instead of using high-end technology like gloves or kinect, we aim to solve this problem using state of the art computer vision and machine learning algorithms

Fig 1.1 American Sign Language Alphabet

Communication is very crucial to human beings, as it enables us to express ourselves. We communicate through speech, gestures, body language, reading, and writing or through visual aids, speech being one of the most commonly used among them. However, unfortunately, for

the speaking and hearing-impaired minority, there is a communication gap. Visual aids, or an interpreter, are used for communicating with them. However, these methods are rather cumbersome and expensive, and can't be used in an emergency. Sign Language chiefly uses manual communication to convey meaning. This involves simultaneously combining hand shapes, orientations and movement of the hands, arms or body to express the speaker's thoughts.

Sign Language consists of fingerspelling, which spells out words character by character, and word level association which involves hand gestures that convey the word meaning. Fingerspelling is a vital tool in sign language, as it enables the communication of names, addresses and other words that do not carry a meaning in word level association. In spite of this, fingerspelling is not widely used as it is challenging to understand and difficult to use. Moreover, there is no universal sign language and very few people know it, which makes it an inadequate alternative for communication.

1.2 Necessity

Communication is one of the basic requirements for survival in society. Deaf and dumb people communicate among themselves using sign language but normal people find it difficult to understand their language. Extensive work has been done on American Sign Language recognition but Indian sign language differs significantly from American Sign Language. ISL uses two hands for communicating (20 out of 26) whereas ASL uses single hand for communicating. Using both hands often leads to obscurity of features due to overlapping of hands. In addition to this, lack of datasets along with variance in sign language with locality has resulted in restrained efforts in ISL gesture detection. Our project aims at taking the basic step in bridging the communication gap between normal people and deaf and dumb people using Indian sign language. Effective extension of this project to words and common expressions may not only make the deaf and dumb people communicate faster and easier with outer world, but also provide a boost in developing autonomous systems for understanding and aiding them.

Deaf people throughout the world, who live in a community with other Deaf people, tend to use signed languages. The particular form of these natural languages varies depending on the context in which they are situated. Signed languages and spoken languages use different modalities. Signed languages are distinct from spoken languages, because any signed language

is articulated by hands, facial expressions, and shoulders and perceived by visual mechanisms. William Stokoe's (Stokoe, Casterline, & Croneberg, 1976) linguistic research led to the realization that ASL has grammatical and syntactical structures and other linguistic features that parallel those found in spoken languages. Ongoing linguistic research continues to design a more fine-grained analysis of the linguistic features of ASL. Due to its visual/gestural nature, ASL has no standard written form, which is also true of signed languages and some spoken languages. Therefore, conventional standards used in some spoken languages for reading and writing are difficult to extrapolate and apply in the same way to the study of ASL, a visual language. Despite the lack of a standard written ASL, forms of ASL literature do exist. Comparable to oral traditions in spoken languages, there is a long-standing use of ASL for storytelling, poetry, drama, and folklore within the Deaf community. The advent of new technology for the preservation and dissemination of language opens new possibilities for personal and artistic expression in ASL

1.3 Objective

The goal of this project was to build a neural network able to classify which letter of the American Sign Language (ASL) alphabet is being signed, given an image of a signing hand. This project is a first step towards building a possible sign language translator, which can take communications in sign language and translate them into written and oral language. Such a translator would greatly lower the barrier for many deaf and mute individuals to be able to better communicate with others in day-to-day interactions.

This goal is further motivated by the isolation that is felt within the deaf community. Loneliness and depression exist in higher rates among the deaf population, especially when they are immersed in a hearing world.

Large barriers that profoundly affect life quality stem from the communication disconnect between the deaf and the hearing. Some examples are information deprivation, limitation of social connections, and difficulty integrating in society.

Most research implementations for this task have used depth maps generated by depth camera and high-resolution images. The objective of this project was to see if neural networks are able to classify signed ASL letters using simple images of hands taken with a personal device such as a laptop webcam. This is in alignment with the motivation as this would make a future

implementation of a real time ASL-to-oral/written language translator practical in an everyday situation.

While automatic speech recognition has now advanced to the point of being commercially available, automatic SLR is still in its infancy. Currently all commercial translation services are human based, and therefore expensive, due to the experienced personnel required

1.4 Theme Of project

SLR aims to develop algorithms and methods to correctly identify a sequence of produced signs and to understand their meaning. Many approaches to SLR in-correctly treat the problem as Gesture Recognition (GR). So, research has thus far focused on identifying optimal features and classification methods to correctly label given sign from a set of possible signs. However, sign language is far more than just a collection of well specified gestures.

Sign languages pose the challenge that they are multi-channel; conveying meaning through many modes at once. While the studies of sign language linguistics are still in their early stages, it is already apparent that this makes many of the techniques used by speech recognition unsuitable for SLR. In addition, publicly avail-able data sets are limited both in quantity and quality, rendering many traditional computers vision learning algorithms inadequate for the task of building classifiers. However, even given the lack of translation tools, most public services are not translated into sign. There is no commonly-used, written form of sign language, so all written communication is in the local spoken language.

The project aims at building a machine learning model that will be able to classify the various hand gestures used for fingerspelling in sign language. In this user independent model, classification machine learning algorithms are trained using a set of image data and testing is done on a completely different set of data. For the image dataset, depth images are used, which gave better results than some of the previous literatures, owing to the reduced pre-processing time. Various machine learning algorithms are applied on the datasets, including Convolutional Neural Network (CNN).

2. LITERATURE REVIEW

2.1 History

Professionals in India believe in an acute shortage of special schools for deaf people. A very few schools use sign language as a medium of instruction. There is also a lack of proper and effective audio-visual support in oral education in these schools. This results in inadequate communication and language skills in the majority of deaf children, impacting on poor literacy skills in the deaf community. The reality is that deaf schools mainly do not use ASL and nearly 5% of deaf people attend deaf schools. The use of ASL is restricted only to vocational programs and short-term courses. ASL was partly influenced by British Sign Language in the finger spelling system and some other signs, but most are unrelated to European sign system.
There was no formal ASL until 1978. The conclusion was that the gestures used in each school were not the same.

It is believed that signing started in India in the 18th century but its use was strongly discouraged. Madan Vasishta sent a questionnaire to the heads of more than hundred schools for the deaf in 1975. Almost all the respondents agreed that there was no ASL. But they also acknowledged that deaf children used some kind of gestures. A similar survey was conducted 20 years later, using a set of questionnaires sent to deaf schools. Some of the responses showed the same misconceptions about sign language that signing is “based on spoken language”, or “based on English”, or “difficult to provide a sign for every spoken word”. Some statements showed that a more positive attitude towards manual communication, and here respondents talked about sign language, rather than gestures. Increasing awareness about the nature of sign languages was verified later on.

Observing the advantages of works on sign language recognition of different countries in aiding the deaf people for communication in public places and accessing/communicating with latest gadgets like Telephone, Computers, etc, linguistic studies on American Sign Language started in 1978 in America. These works resulted in ASL and discovered that it is a language on its own with specific syntax, grammar, phonology and morphology. Static numerals of ASL isolated gestures are shown in Figure-1.

Professionals in America believe in an acute shortage of special schools for deaf people. A very few schools use sign language as a medium of instruction. There is also a lack of proper and effective audio-visual support in oral education in these schools. This results in inadequate Communication and language skills in the majority of deaf children, impacting on poor literacy skills in the deaf community.

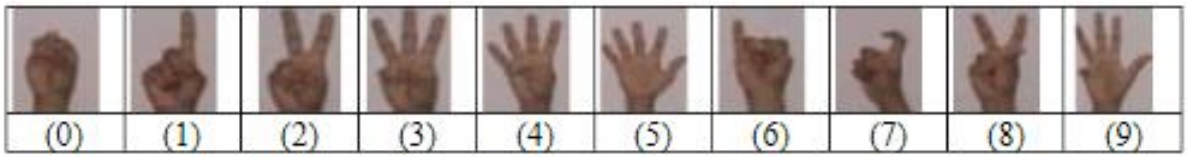


Fig 2.1. American Sign Language (Numbers)

While significant progress has already been made in computer recognition of sign languages of other countries but a very limited work has been done in ASL computerization. The works carried by various researchers worldwide are summarized in this paper. The domain is isolated sign language but continuous sign language recognition is also discussed due to similarity with isolated sign language recognition. We also select research papers where no special image acquiring devices are required. The reason is that in common places no special image acquiring devices are available at all the times, and all deaf/mute/hard hearing persons might be unable to wear due to their economic conditions and in most cases, it is cumbersome to carry and wear. Also, we select few research papers in which special wearable devices are used as inputs due to their better performance for comparison purposes.

2.2 Existing Project and Research Paper

2.2.1 Sign Language Recognition

- Publication Date: Oct’21
- Author
    - Veenu U
    - Dennis G Dayakant
    - Shwetha S

- Publication:

  Research Gate

➢ **Brief Introduction**

Sign Language is a language which uses gestures, where dumb and deaf people interact with the common people or people of their community by performing some actions or gestures. At times or not all of the common people are familiar with the AS Language, hence there is a communication barrier between them. In order to break this communication-barrier we have developed a model which recognizes the gestures of the dumb and deaf people and translate them to the common people in their understandable language. In order to help the mute, we have built this model. Let's take an example of a subject who had a stroke and had lost their speech. It would mentally disturb them when they realise, they can no more talk and every time they try to communicate the other individual would not understand. But what if there was an app that would detect the sign language and transfer it to text which makes communication much easier.

This model comes in handy. It's a perceptive system that can predict and anticipate sign language. Deep learning has aided in the development of new algorithms and architectures that support artificial neural networks, such as the SSD MobileNet V2 algorithm. Different types of sign language exist in this world some of them are ASL (American Sign Language), GSL (German Sign Language), ISL (Indian Sign Language), BSL (British Sign Language), and so on. Sign Language communication is multifaceted, it uses only hand gestures, body movements and facial expressions. This paper is mainly focusing on the hand gestures. This representation will help the computer to distinguish between possible classes of actions. In our work extraction is automated using TensorFlow.

➢ **Methodology Used**

First, we collect images using python and OpenCV. Next, we label them using label image package. We then move into building the sign language detector using Transfer learning and TensorFlow object detection API. We use SSD mobile net v2 to help detect the images.

Object Detection using SSD MOBILENET algorithm. It is possible to achieve a very good balance between accuracy and speed when using the Single Shot Detector. SSD computes

a feature map by running a convolutional neural network on a given input image only once. In SSD, a small 3x3-sized convolutional kernel is applied to this feature map to predict the bounding boxes and classification probability. It also employs anchor boxes with varying aspect ratios, similar to Faster-RCNN, and learns the off-set rather than the box. After multiple convolutional layers, the SSD predicts bounding boxes. Because each convolutional layer operates at a different scale, it can detect objects of varying sizes.

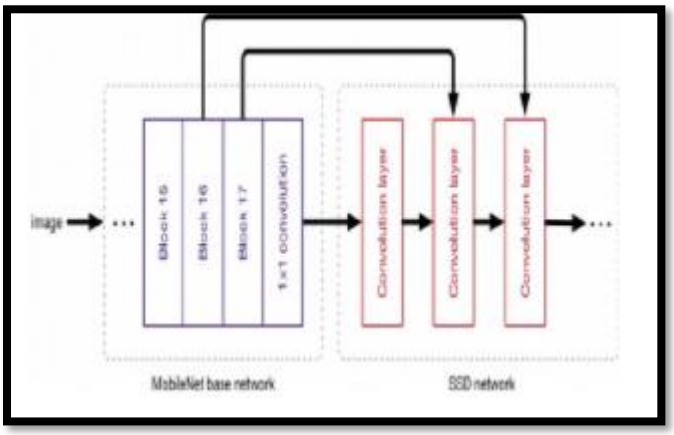


Fig 2.2 SSD MobileNet V2

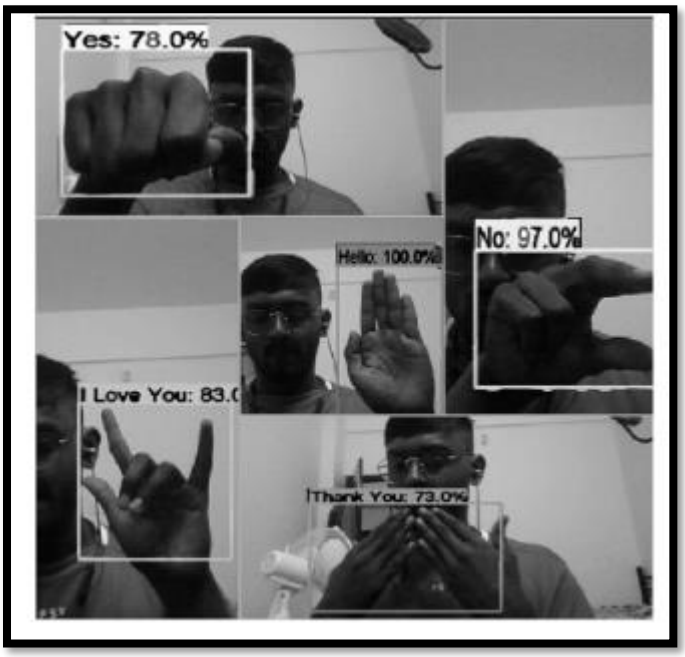


Fig 2.3. Process of Object Detection

2.2.2 Sign Language Recognition System using Convolutional Neural Network and Computer Vision

- Publication date: Dec'20
- Author:
  - Mehreen Hurroo
  - Mohammad Elham
- Publication:

  IJERT

- **Brief Introduction:**

As well stipulated by Nelson Mandela, talk to a man in a language he understands, that goes to his head. Talk to him in his own language that goes to his heart, language is undoubtedly essential to human interaction and has existed since human civilisation began. It is a medium humans use to communicate to express themselves and understand notions of the real world. Without it, no books, no cell phones and definitely not any word I am writing would have any meaning. It is so deeply embedded in our everyday routine that we often take it for granted and don't realise its importance. Sadly, in the fast-changing society we live in, people with hearing impairment are usually forgotten and left out. They have to struggle to bring up their ideas, voice out their opinions and express themselves to people who are different to them. Sign language, although being a medium of communication to deaf people, still have no meaning when conveyed to a non-sign language user. Hence, broadening the communication gap. To prevent this from happening, we are putting forward a sign language recognition system. It will be an ultimate tool for people with hearing disability to communicate their thoughts as well as a very good interpretation for non-sign language user to understand what the latter is saying. Many countries have their own standard and interpretation of sign gestures. For instance, an alphabet in Korean sign language will not mean the same thing as in Indian sign language. While this highlights diversity, it also pinpoints the complexity of sign languages. Deep learning must be well versed with the gestures so that we can get a decent accuracy. In our proposed system, American Sign Language is used to create our datasets.

Identification of sign gesture is performed with either of the two methods. First is a glove-based method whereby the signer wears a pair of data gloves during the capture of hand

movements. Second is a vision-based method, further classified into static and dynamic recognition. Static deals with the 2dimensional representation of gestures while dynamic is a real time live capture of the gestures.

And despite having an accuracy of over 90%, wearing of gloves are uncomfortable and cannot be utilised in rainy weather. They are not easily carried around since their use require computer as well. In this case, we have decided to go with the static recognition of hand gestures because it increases accuracy as compared to when including dynamic hand gestures like for the alphabets J and Z. We are proposing this research so we can improve on accuracy using Convolution Neural Network (CNN).

- **Methodology Used**

The first step of the proposed system is to collect data. Many researchers have used sensors or cameras to capture the hand movements. For our system, we make use of the web camera to shoot the hand gestures. The images undergo a series of processing operations whereby the backgrounds are detected and eliminated using the colour extraction algorithm HSV (Hue Saturation Value). Segmentation is then performed to detect the region of the skin tone. Using the morphological operations, a mask is applied on the images and a series of dilation and erosion using elliptical kernel are executed. With OpenCV, the images obtained are amended to the same size so there is no difference between images of different gestures. Our dataset has 2000 American sign gesture images out of which 1600 images are for training and the rest 400 are for testing purposes. It is in the ratio 80:20. Binary pixels are extracted from each frame, and Convolutional Neural Network is applied for training and classification. The model is then evaluated and the system would then be able to predict the alphabets.

Data Collection Image capture: web camera

Image Processing

1. Backgrounds detected and eliminated with HSV
2. Morphological operations are performed and masks are applied.
3. Segmentation of hand gesture
4. Image is resized

Feature Extraction

- Binary pixels
- Classification
- Evaluation: The precision, recall and F- measures for each class are determined.

Using CNN of 3 layers:

1. Three Convolutional layers with ReLU as activation function
2. Three layers of pooling
3. Fully connected layer with Softmax as activation function

Prediction: The system predicts input gesture of user and displays result.

2.2.3 Sign Language Recognition Application Systems for Deaf-Mute People: A Review Based on Input-Process-Output

- Publication date: Oct'17
- Author:
  - Suharjito
  - Ricky Anderson
  - Fanny Wiryana
  - Meita ChandraAriesta
  - Gede PutraKusuma
- Publication:
  ScienceDirect

➢ **Brief Introduction:**

Deaf is a disability that impair their hearing and make them unable to hear, while mute is a disability that impair their speaking and make them unable to speak. Both are only disabled at their hearing and/or speaking, therefore can still do much other things. The only thing that separates them and the normal people is communication. If there is a way for normal people and deaf-mute people to communicate, the deaf-mute people can easily live like a normal person. And the only way for them to communicate is through sign language. While

sign language is very important to deaf-mute people, to communicate both with normal people and with themselves, is still getting little attention from the normal people. We as the normal people, tend to ignore the importance of sign language, unless there are loved ones who are deaf-mute. One of the solutions to communicate with the deaf-mute people is by using the services of sign language interpreter. But the usage of sign language interpreter can be costly. Cheap solution is required so that the deaf-mute and normal people can communicate normally. Therefore, researchers want to find a way for the deaf-mute people so that they can communicate easily with normal person.

The breakthrough for this is the Sign Language Recognition System. The system aims to recognize the sign language, and translate it to the local language via text or speech. However, building this system cost very much and are difficult to be applied for daily use. Early researches have known to be successful in Sign Language Recognition System by using data gloves. But, the high cost of the gloves and wearable character make it difficult to be commercialized. Knowing that, researchers then try to develop a pure vision Sign Language Recognition Systems. However, it is also coming with difficulties, especially to precisely track hands movements. The problems of developing sign language recognition ranges from the image acquisition to the classification process. Researchers are still finding the best method for the image acquisition. Gathering images using camera gives the difficulties of image pre-processing.

Meanwhile, using active sensor device can be costly. Classification methods also give researchers some drawbacks. Wide choice of recognition method makes researchers unable to focus on one best method. Choosing one method to be focused on, tends to make other method that may be better suit for Sign Language Recognition, not being tested. Trying out other methods makes researchers barely develops one method to its fullest potentials. This paper aims to discuss the Sign Language Recognition System that are being used by researchers. In this paper, we will discuss about the Sign Language Recognition form application point of view. This paper will talk about the device used in getting the data, data acquisition, such as data from early researches or self-made data, the recognition method that are recently used by researchers, and the output of previous researches.

- **Methodology Used**

Processing Methods One of the processing methods widely used in SLR is the Hidden Markov Model (HMM). 18 uses the HMM algorithm by combining it with another algorithm. Firstly, they use fuzzy K-means algorithm to do the Hand Shape Classifier. The hand shapes are classified to 8 clusters for the right hand and 7 clusters for the left hand (based on the algorithm testing). And then, the orientation classifier is done by classifying the direction which the palm is facing. Due to large variety of SLR gesture, the classification is done by classifying the onset and offset of hand orientation, as it is relatively consistent when same sign is done repeatedly. Lastly, they do the movement classifier by using HMM.

The HMM they use is Multi Stream HMM (MSHMM). It is based on two standard single-stream HMM. The ACC and sEMG data feature sets are used to train the stream via Baum-Welch algorithm. The other method of HMM is used by 3 which is called Light-HMM. They select the key frames through low-rank approximation and adaptively determine the hidden states in HMM. By using such method, the key frames can be reduced and the estimation of hidden states can be more precise. The basic idea is that the frames with linearly independent features will be selected as key frames and the others will be discarded. There are also a HMM approach that is called TD-HMM 14. This method uses the method similar to continuous HMM (CHMM), with the amount of computation is reduced by tying similar Gaussian mixture components.

The other processing method is neural network. 10 uses neural network called 3D Convolutional Neural Network (CNN). The 3D CNN method is based on 2D CNN which is used in speech and image recognition. The 2D CNN implementation is done by extracting a layer to feature map by using kernel window called local receptive field. It reduces the free variables and increase generalization capability of the network. The 3D CNN is implemented using 2D CNN by adding the motion feature. The 3D Convolution is achieved by convolving 3D kernel with the cube formed by stacking multiple neighbouring frames together. 12 uses two 3D-CNN in their research, for extracting hand and upper body features. Each CNN is 3 layers deep. They also applied local contrast normalization (LCN) for the first two layer and the neurons are rectified linear units (ReLU). 21 uses 2D

Convolutional Neural Network in their research, with unusual feature in their network architecture.

The unusual features are ReLU Nonlinearity, multiple GPUs, Local Response Normalization, and Overlapping Pooling. The ReLU Nonlinearity is the alternative of traditional neuron models in CNN. It trains several times faster than traditional CNN and is best fit to be used in a big dataset. They also use multiple GPUs to train their data be cause of the size of data that do not fit in one GPU. While ReLU on its own does not need normalization, they still add the Local Response Normalization to their network. The normalization aids in generalization and it decreases the test error rates. They also use the overlapping method, differ from traditional CNN that does not overlap, and find it more difficult to cause overfitting. They also use Data Augmentation and Dropout method that reduces overfitting. The dropout method is a new method that drops neurons so that the network is forced to learn more robust feature.

Authors	Method or Application Systems	Accuracy Result
[3]	Light-HMM	83,6%
[18]	Multi Stream HMM	86,7%
[10]	3D-CNN	94,2%
[12]	3D-CNN	78,8%

Fig 2.4. Comparison result of Sign Language Recognition

3. SYSTEM DEVELOPMENT

3.1 Requirement Specification

➢ Python

Python is an interpreted high-level general-purpose programming language. Its design philosophy emphasizes code readability with its use of significant indentation. Its language constructs as well as its object-oriented approach aim to help programmers write clear, logical code for small and large-scale projects.

- Developer: Python Software Foundation
- First appeared: 20 February 1991; 30 years ago
- Designed by: Guido van Rossum
- OS: Windows, Linux/UNIX, macOS and more
- Paradigm: Multi-paradigm: object-oriented, procedural (imperative), functional, structured, reflective
- Stable release: 3.10.1 / 6 December 2021

➢ Jupyter

The Jupyter Notebook is an open-source web application that you can use to create and share documents that contain live code, equations, visualizations, and text. ... The name, Jupyter, comes from the core supported programming languages that it supports: Julia, Python, and R

Project Jupyter is a project and community whose goal is to "develop open-source software, open-standards, and services for interactive computing across dozens of programming languages". It was spun off from IPython in 2014 by Fernando Pérez and Brian Granger.

- Purpose: To support interactive data science and scientific computing across all programming languages.
- Founded: February 2015
- Abbreviation: Jupyter
- Founders: Fernando Pérez, Brian Granger

- **Keras**

Keras is an open-source software library that provides a Python interface for artificial neural networks. Keras acts as an interface for the TensorFlow library. ... Designed to enable fast experimentation with deep neural networks, it focuses on being user-friendly, modular, and extensible.

- Repository: github.com/keras-team/keras
- Written in: Python
- Platform: Cross-platform
- License: MIT
- Initial release: 27 March 2015
- Stable release: 2.4.0 / 17 June

- **OpenCV**

OpenCV is a library of programming functions mainly aimed at real-time computer vision. Originally developed by Intel, it was later supported by Willow Garage then Itseez. The library is cross-platform and free for use under the open-source Apache 2 License. Wikipedia

- Size: ~200 MB
- License: Apache license
- Stable release: 4.5.4 / 9 October 2021
- Written in: C/C++
- Original author(s): Intel Corporation, Willow Garage, Itseez
- Initial release: June 2000

- **CNN**

In deep learning, a convolutional neural network (CNN, or ConvNet) is a class of artificial neural network, most commonly applied to analyse visual imagery. They are also known as shift invariant or space invariant artificial neural networks (SIANN), based on the shared-weight architecture of the convolution kernels or filters that slide along input features and provide translation equivariant responses known as feature

maps. Counter-intuitively, most convolutional neural networks are only equivariant, as opposed to invariant, to translation. They have applications in image and video recognition, recommender systems, image classification, image segmentation, medical image analysis, natural language processing, brain-computer interfaces, and financial time series.

- Convolution Neural Networks or covnets are neural networks that share their parameters. Imagine you have an image. It can be represented as a cuboid having its length, width (dimension of the image), and height (as images generally have red, green, and blue channels).

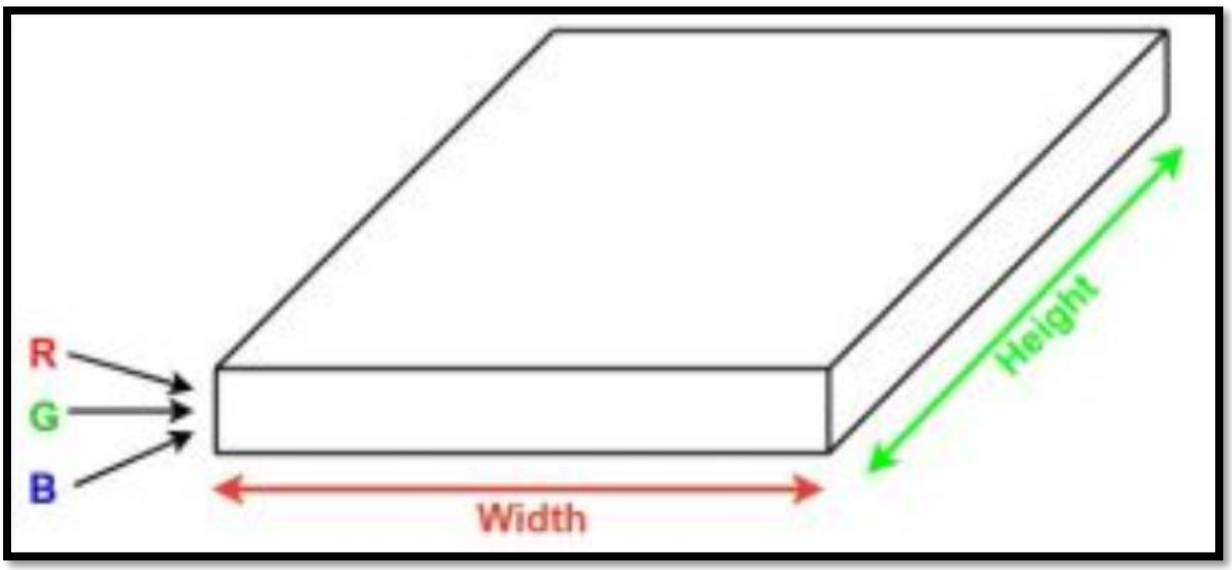


Fig 3.1 Representation of Cuboid

- Now imagine taking a small patch of this image and running a small neural network on it, with say, k outputs and represent them vertically. Now slide that neural network across the whole image, as a result, we will get another image with different width, height, and depth. Instead of just R, G, and B channels now we have more channels but lesser width and height. This operation is called Convolution. If the patch size is the same as that of the image it will be a regular neural network. Because of this small patch, we have fewer weights.

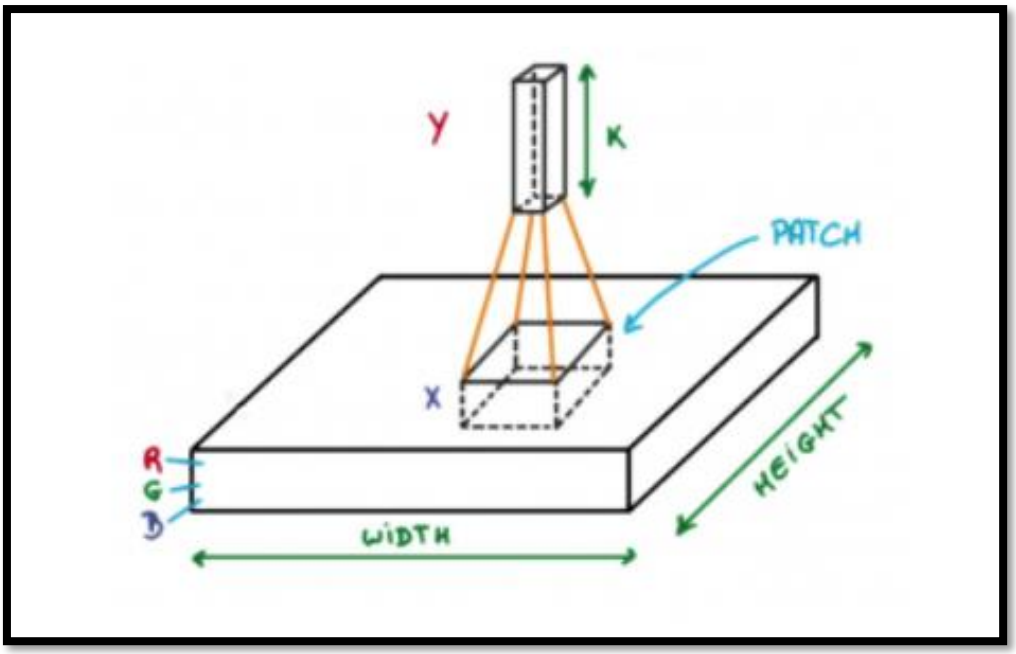


Fig 3.2 Understanding CNN layers

- Now let's talk about a bit of mathematics that is involved in the whole convolution process.
    - Convolution layers consist of a set of learnable filters (a patch in the above image). Every filter has small width and height and the same depth as that of input volume (3 if the input layer is image input).
    - Convolution layers consist of a set of learnable filters (a patch in the above image). Every filter has small width and height and the same depth as that of input volume (3 if the input layer is image input).
    - Convolution layers consist of a set of learnable filters (a patch in the above image). Every filter has small width and height and the same depth as that of input volume (3 if the input layer is image input).
    - Convolution layers consist of a set of learnable filters (a patch in the above image). Every filter has small width and height and the same depth as that of input volume (3 if the input layer is image input).

- Layers used to build ConvNets: A covnets is a sequence of layers, and every layer transforms one volume to another through a differentiable function.

- Types of layers: Let's take an example by running a covnets on of image of dimension 32 x 32 x 3.
    - Input Layer: This layer holds the raw input of the image with width 32, height 32, and depth 3.
    - Convolution Layer: This layer computes the output volume by computing the dot product between all filters and image patches. Suppose we use a total of 12 filters for this layer we'll get output volume of dimension 32 x 32 x 12.
    - Activation Function Layer: This layer will apply an element-wise activation function to the output of the convolution layer. Some common activation functions are RELU: max(0, x), Sigmoid: 1/(1+e^-x), Tanh, Leaky RELU, etc. The volume remains unchanged hence output volume will have dimension 32 x 32 x 12.
    - Pool Layer: This layer is periodically inserted in the covnets and its main function is to reduce the size of volume which makes the computation fast reduces memory and also prevents overfitting. Two common types of pooling layers are max pooling and average pooling. If we use a max pool with 2 x 2 filters and stride 2, the resultant volume will be of dimension 16x16x12.

- How does it work?
    - Generally, A Convolutional neural network has three layers. And we understand each layer one by one with the help of an example of the classifier. With it can classify an image of an X and O. So, with the case, we will understand all four layers.
    - Convolutional Neural Networks have the following layers:
        - Convolutional
        - ReLU Layer
        - Pooling
        - Fully Connected Layer

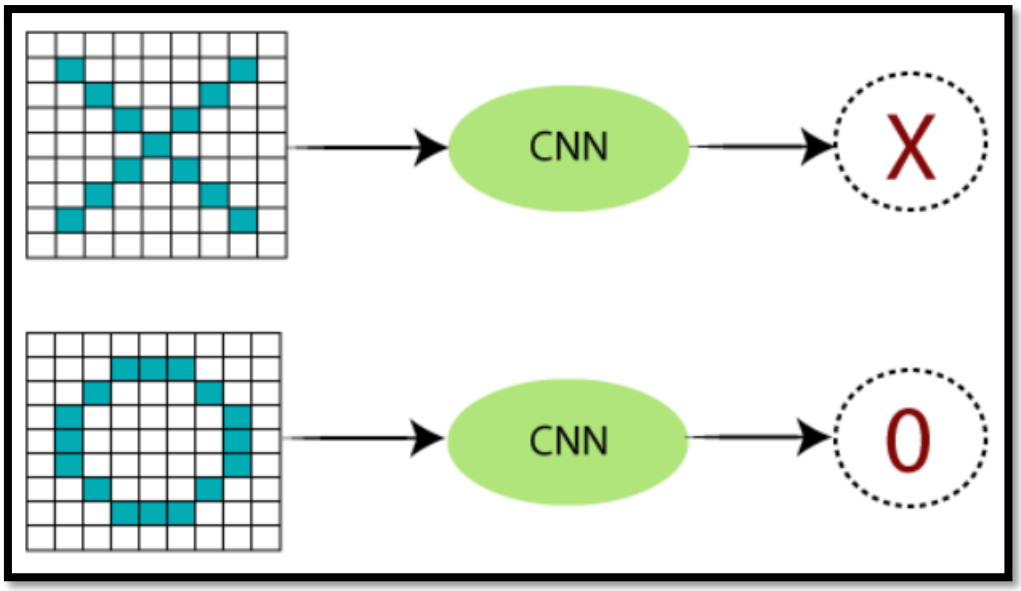


Fig 3.3 Classifying CNN

- There are certain trickier cases where X can represent in these four forms as well as the right side, so these are nothing but the effects of the deformed images. Here, there are multiple presentations of X and O's. This makes it tricky for the computer to recognize. But the goal is that if the input signal looks like previous images, it has seen before, the "image" reference signal will be convolved with, the input signal. The resulting output signal is then passed on to the next layer. Consider the diagram shown below:

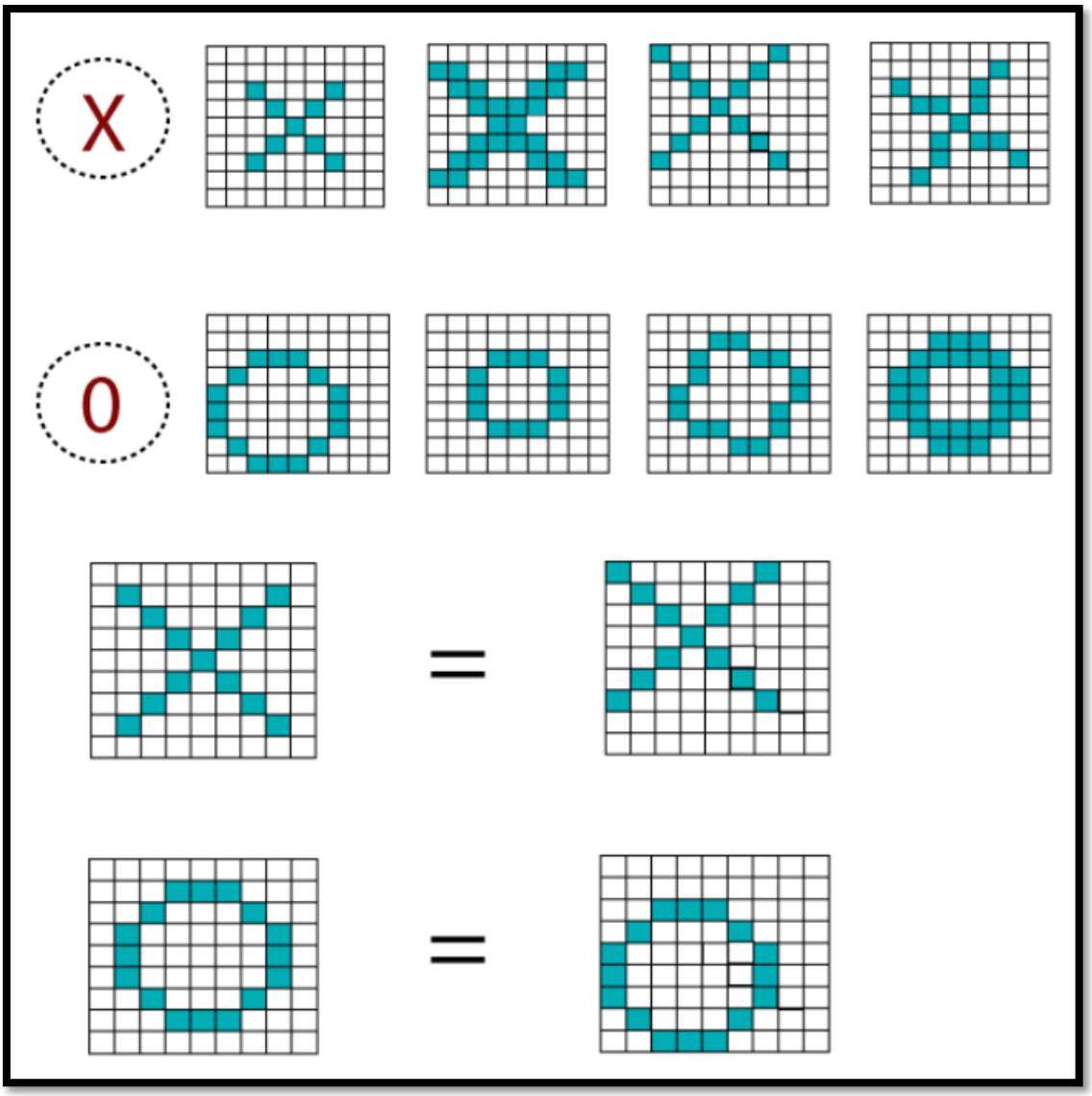


Fig 3.4 Example of CNN

- A computer understands an image using numbers at each pixel.

➢ **TensorFlow**

TensorFlow is a free and open-source software library for machine learning and artificial intelligence. It can be used across a range of tasks but has a particular focus on training and inference of deep neural networks. Wikipedia

- Platform: Linux, macOS, Windows, Android, JavaScript
- License: Apache License 2.0
- Developer(s): Google Brain Team

- Initial release: November 9, 2015
- Stable release: 2.6.1 (1 November 2021) / May 14, 2021
- Programming languages: Python, C++

➢ **H5py**

The h5py package is a Pythonic interface to the HDF5 binary data format. HDF5 lets you store huge amounts of numerical data, and easily manipulate that data from NumPy. For example, you can slice into multi-terabyte datasets stored on disk, as if they were real NumPy arrays.

3.1.1 Data Flow Diagram (DFD)

Level 0 DFD

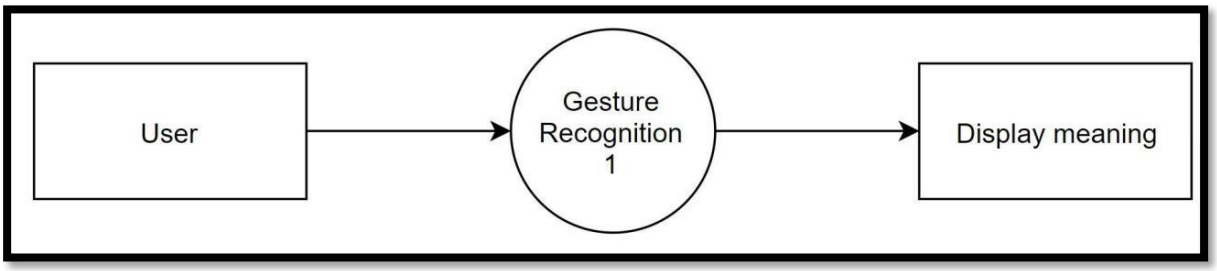


Fig 3.5. Level 0 DFD

In this level 0 data flow diagram, the whole system is represented with the help of input, processing and output. The input to the sign recognition system is the live feed from the camera which contains the gestures performed by the user. The camera provides the frames which can be mapped to their corresponding meanings.

Level 1 DFD

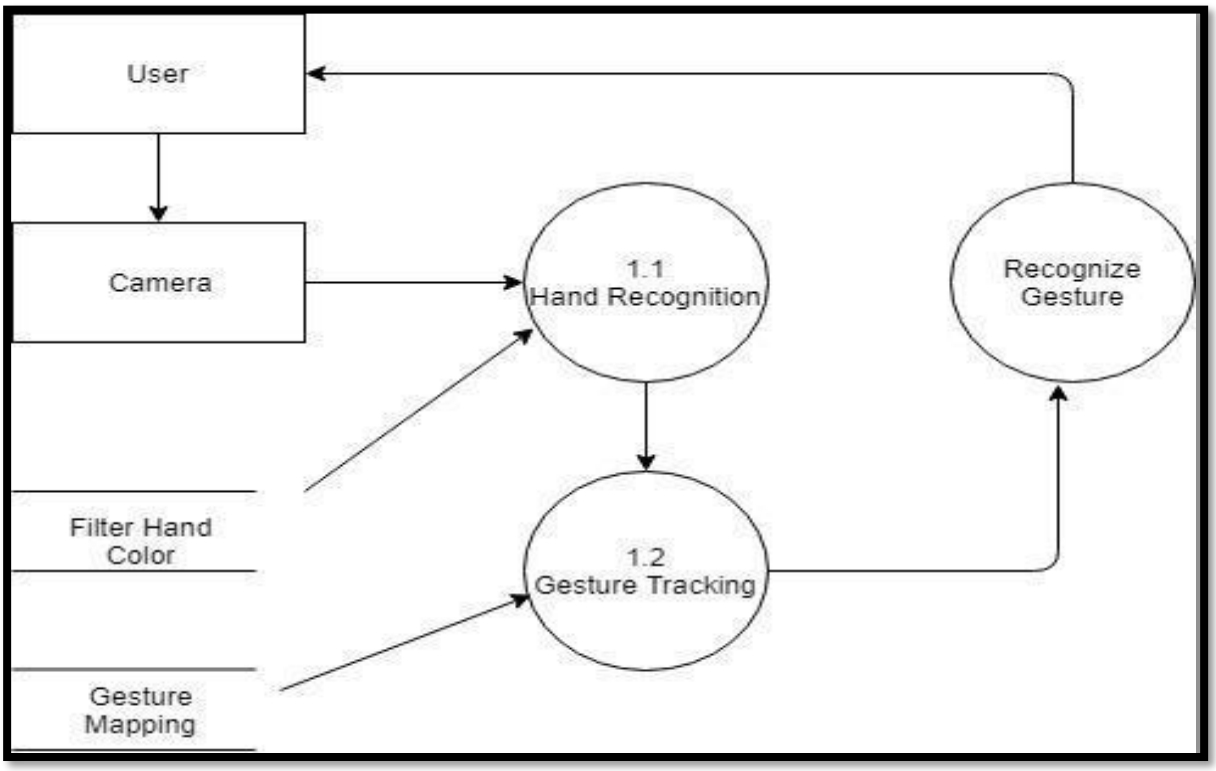


Fig 3.6. Level 1 DFD

In level 1 data flow diagram, the gesture recognition module is explained in further detail. The camera provides live feed of the user actions. Operations are performed to enhance the hand movements. The hand movement is recognized and sent to the gesture tracking module. The gesture tracking module checks for the gesture in the pre trained network of gestures and their respective meanings. The gesture control module maps the gesture to its meaning. Thus, in these processes the gesture is recognized and meaning is displayed.

Level 2 DFD

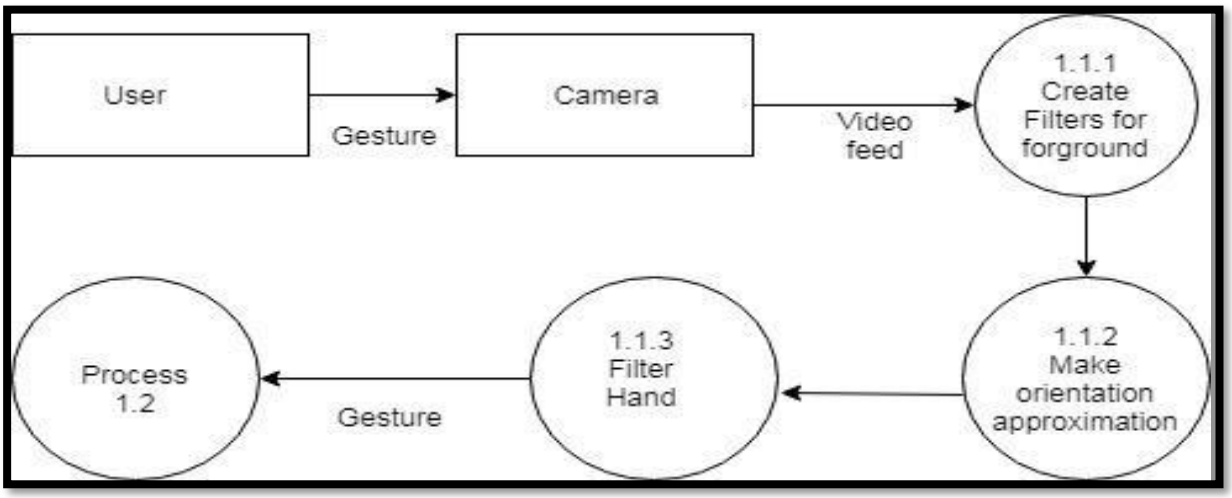


Fig 3.7.a Level 2 DFD

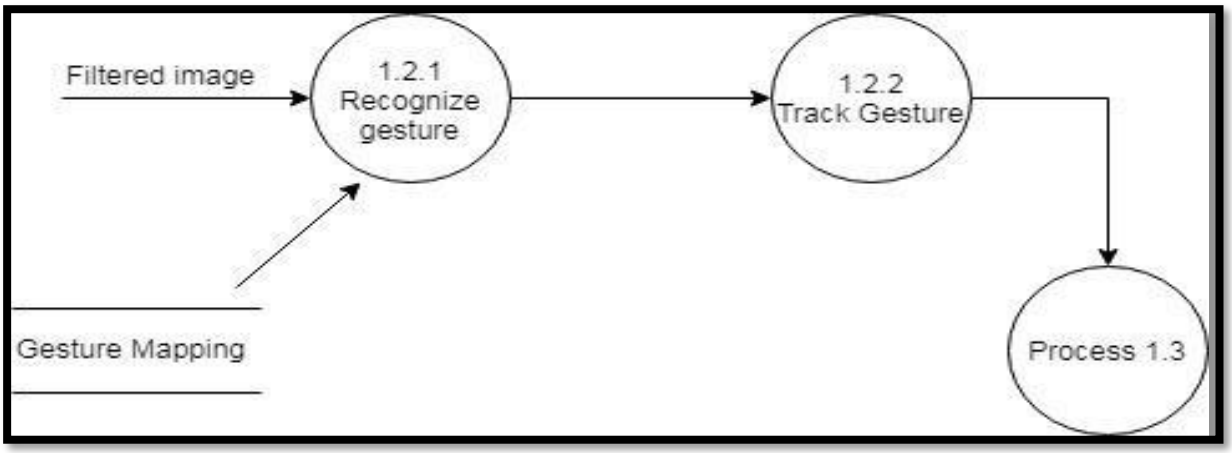


Fig 3.7.b Level 2 DFD

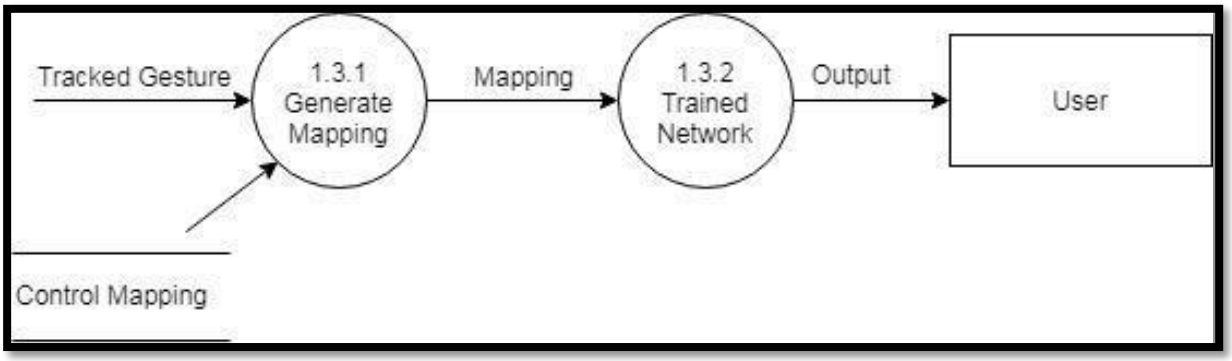


Fig 3.7.c Level 2 DFD

In the level 2 data flow diagram, the three important processes are explained:

Process 1.1 - here the video is captured via the camera and sent for processing. Convolution operations are performed on the feed to enhance CNN filters. The primary focus is the hand of the user. The hand is recognized by training CNN over a number of samples.

Process 1.2 - here the filtered images are stitched back into a video to generate RNN data using pool features of CNN, the gestures are recognized and tracking takes place.

Process 1.3 - the tracked gesture is obtained and mapped to its meaning via the pre trained network.

3.1.2 Specification Document/ UML Diagrams

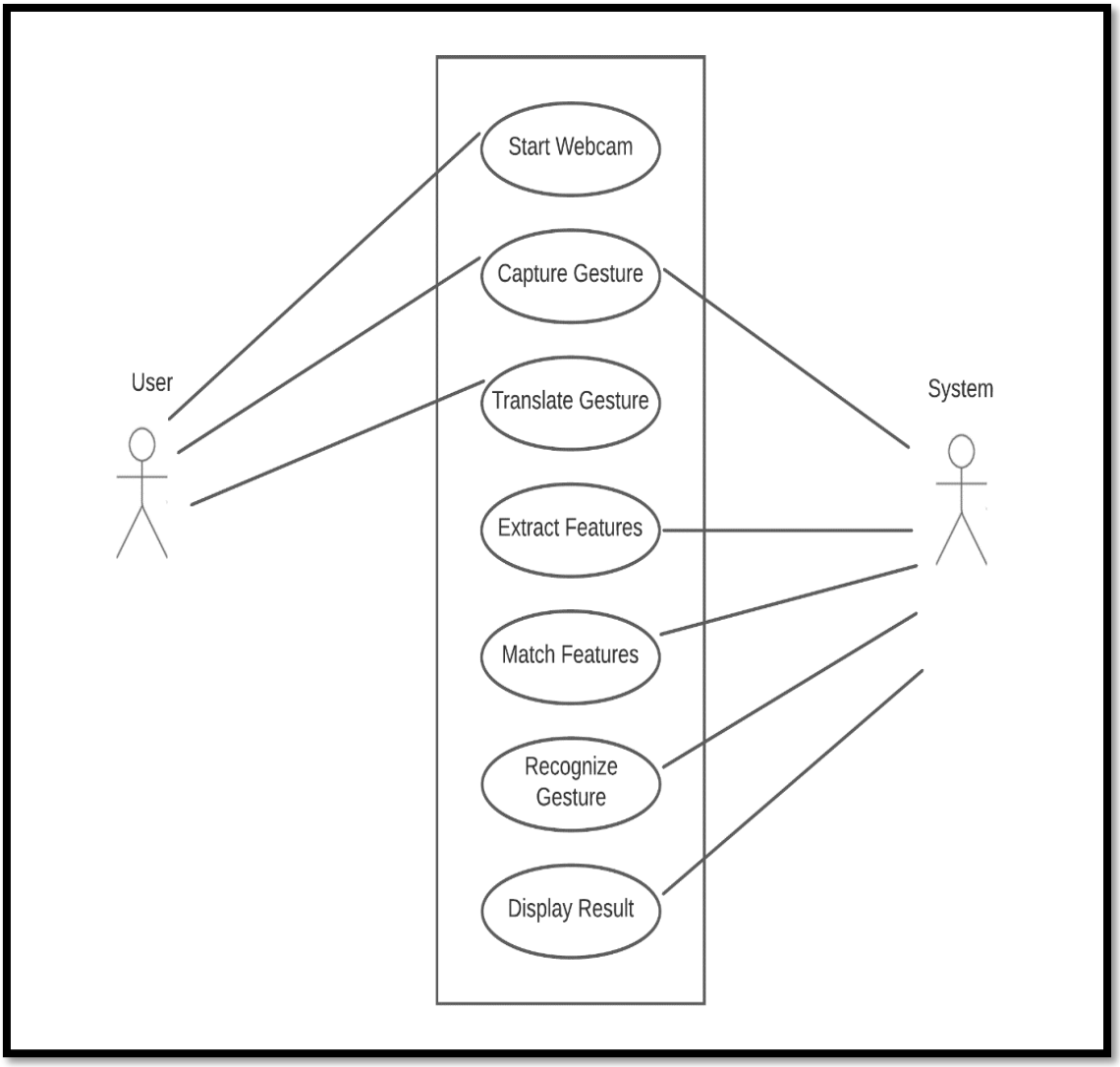


Fig 3.8. Use Case Diagram for Sign language recognition

Above Figure represent Use Case Diagram for Sign language recognition. With reference to the figure as we can see user have to start the webcam in order to capture the gestures of hands. So, as it can be translated and give the input feed to the system.

The system will extract and store the extracted gesture and will try to match the feature present in the dataset so it can recognize the gesture and it will display the output we reference to the captured gesture.

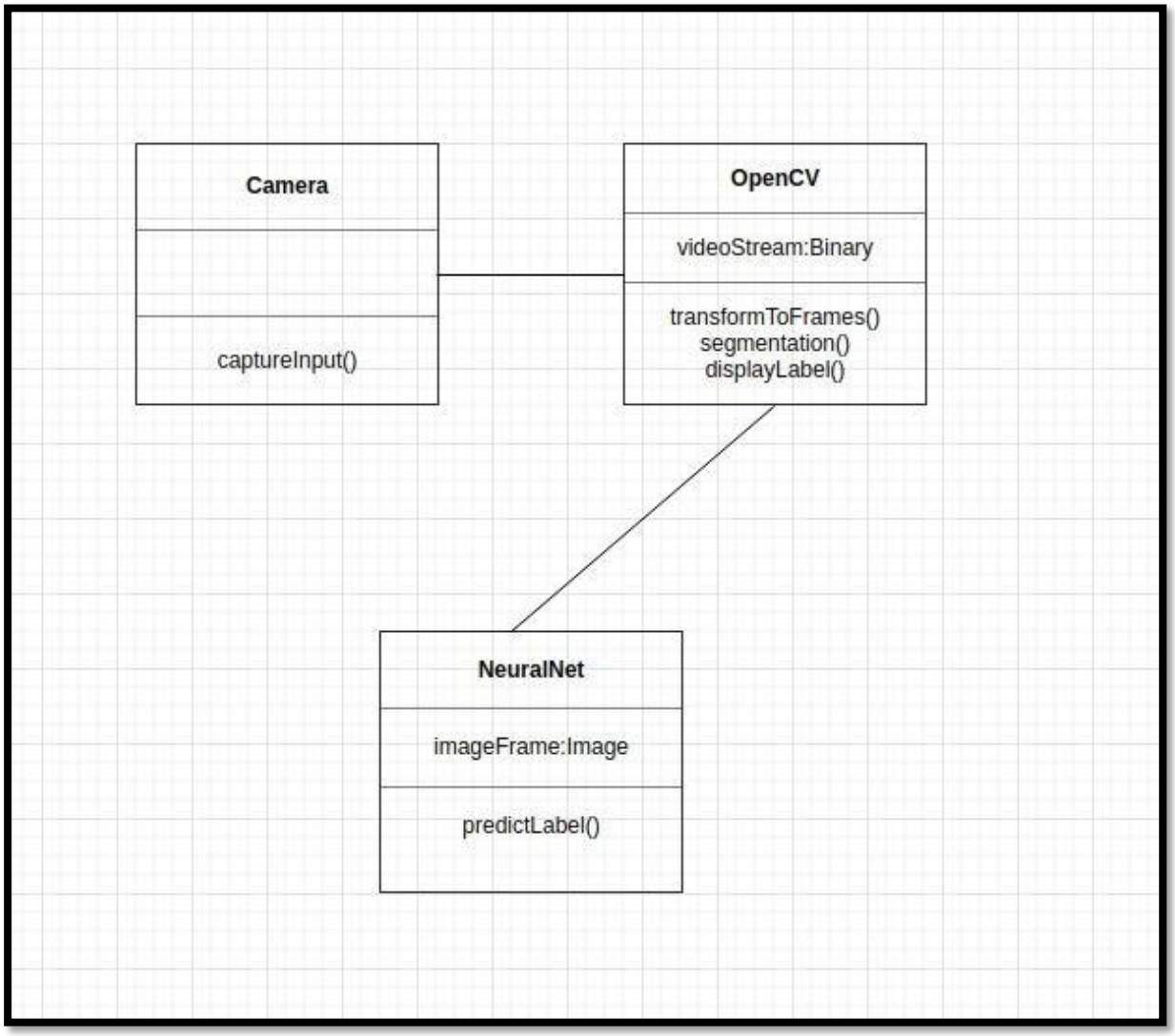


Fig 3.9. Class Diagram of Sign Language Recognition

As in the given figure we can see that the input is capture through the function name captureInput() via the camera. Followed by storing the gesture as well as extracting the features

from given input. After that with the help of openCv library the input image is transformed into the frames by using tranformToFrames() function.

The Image is segmented into different segment by using segmentation() function and the labels are displayed via displayLabel() Function.

Next, the dataset is trained by neural network model in which the labels are predicted and the matching output is displayed through the system on the screen.

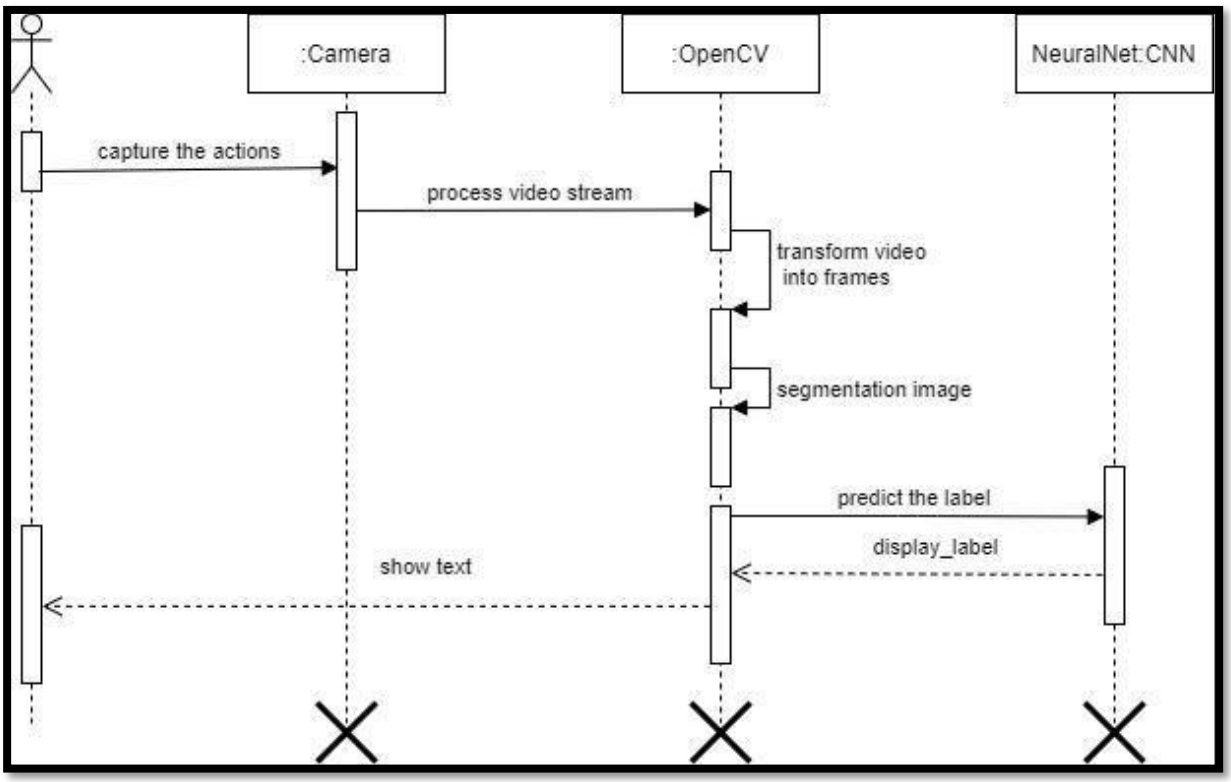


Fig 3.10. Sequence Diagram of Sign language recognition

In the above diagram we can see the sequence of the project named as Sign Language Recognition which start with capturing the action through the camera of the hand, followed by processing the video stream with the help of OpenCV library. The video is transformed into the frames, and segmentation of the images is done so as it is easy for the system to classify the image input and is send to CNN for predicting. After predicting the label, it is displayed again on the screen and the output is the form of text.

3.2 User Interface Design

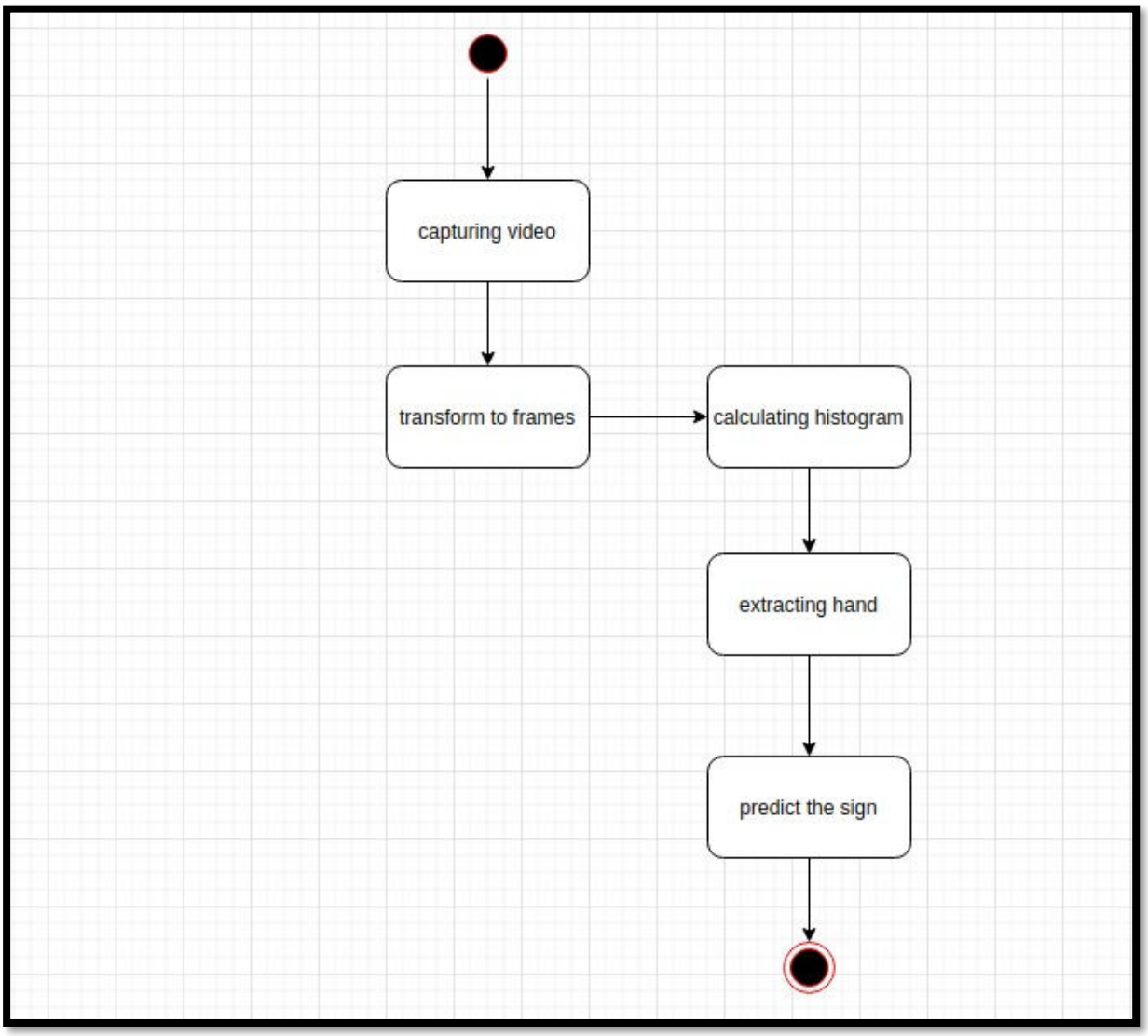


Fig 3.11. State Chart Diagram

According to the above figure, we can see that in the state diagram we first take the input from the camera and start to capture the video for getting hand gestures.

The video is then transformed to frames where the calculation of the histogram is done and the features of hand gestures are extracted by using OpenCV. Then the prediction is done by CNN model. After that the output is displayed on the screen

3.2 Database Design / ER Diagrams

3.3.1 Design

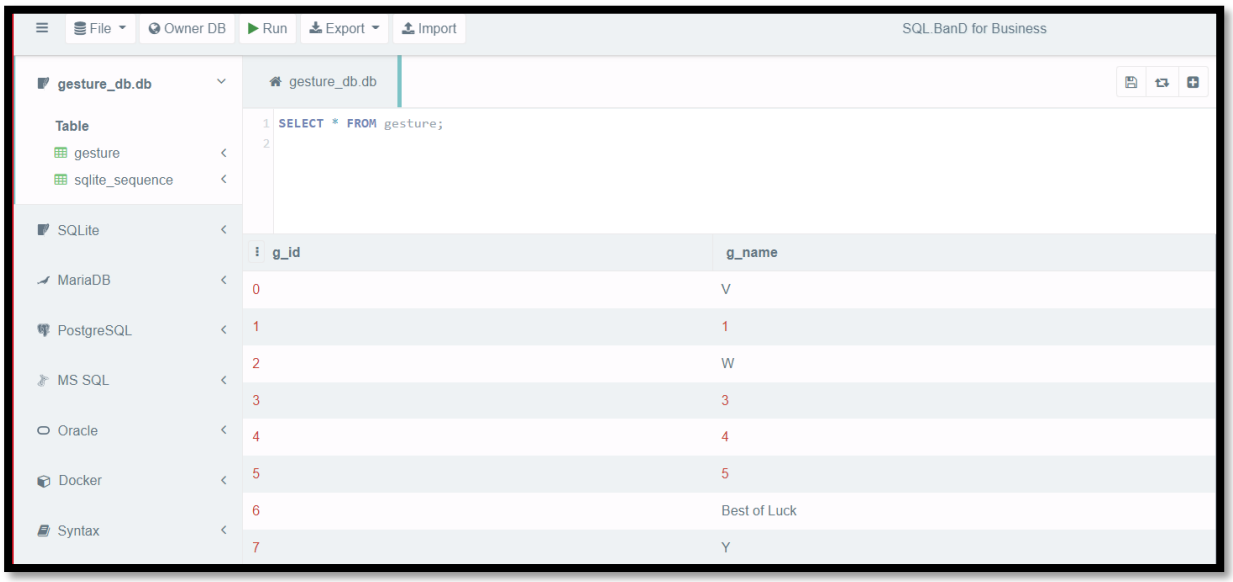


Fig 3.12. Database Design

In the above Database Design as we can see we have created the database for our project in MySQL. In the database we have taken gestures in the form of Numbers, Letters, and Abbreviation.

Which has g_id i.e., gesture id, g_name i.e gesture name. This database is used to extract the g_id when the CNN model predicts the text label so that in order it will display the gesture name associated with its gesture id.

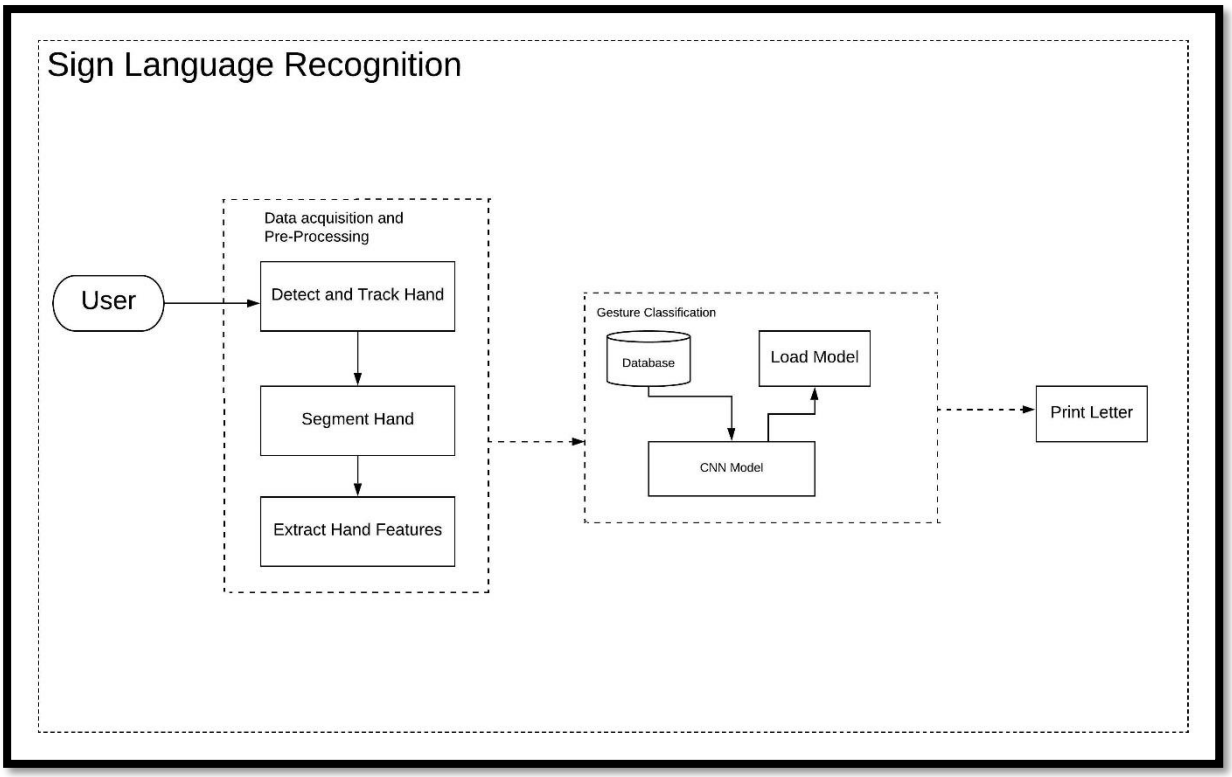


Fig 3.13. ER Diagram

3.3.2 Diagram

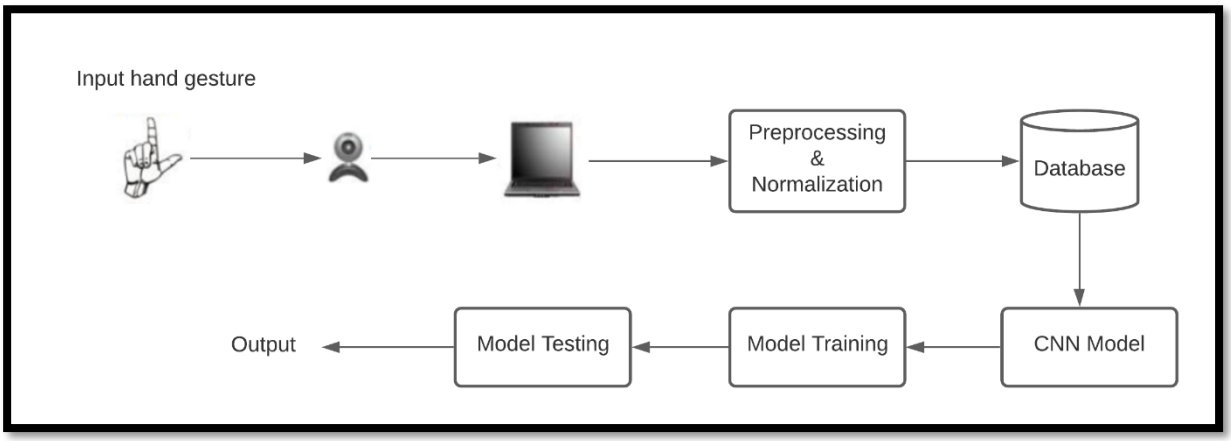


Fig 3.14. Horizontal flowchart

3.3.3 Database Diagram

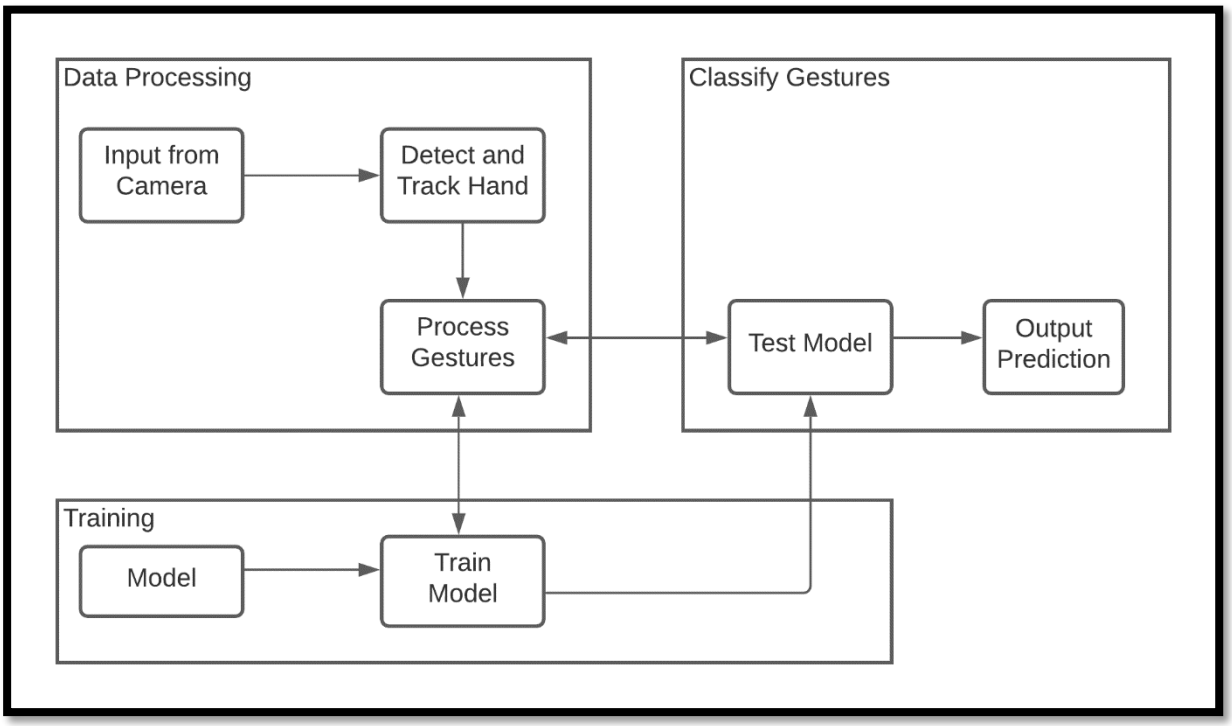


Fig 3.15. Database Diagram

3.3.4 Flowchart

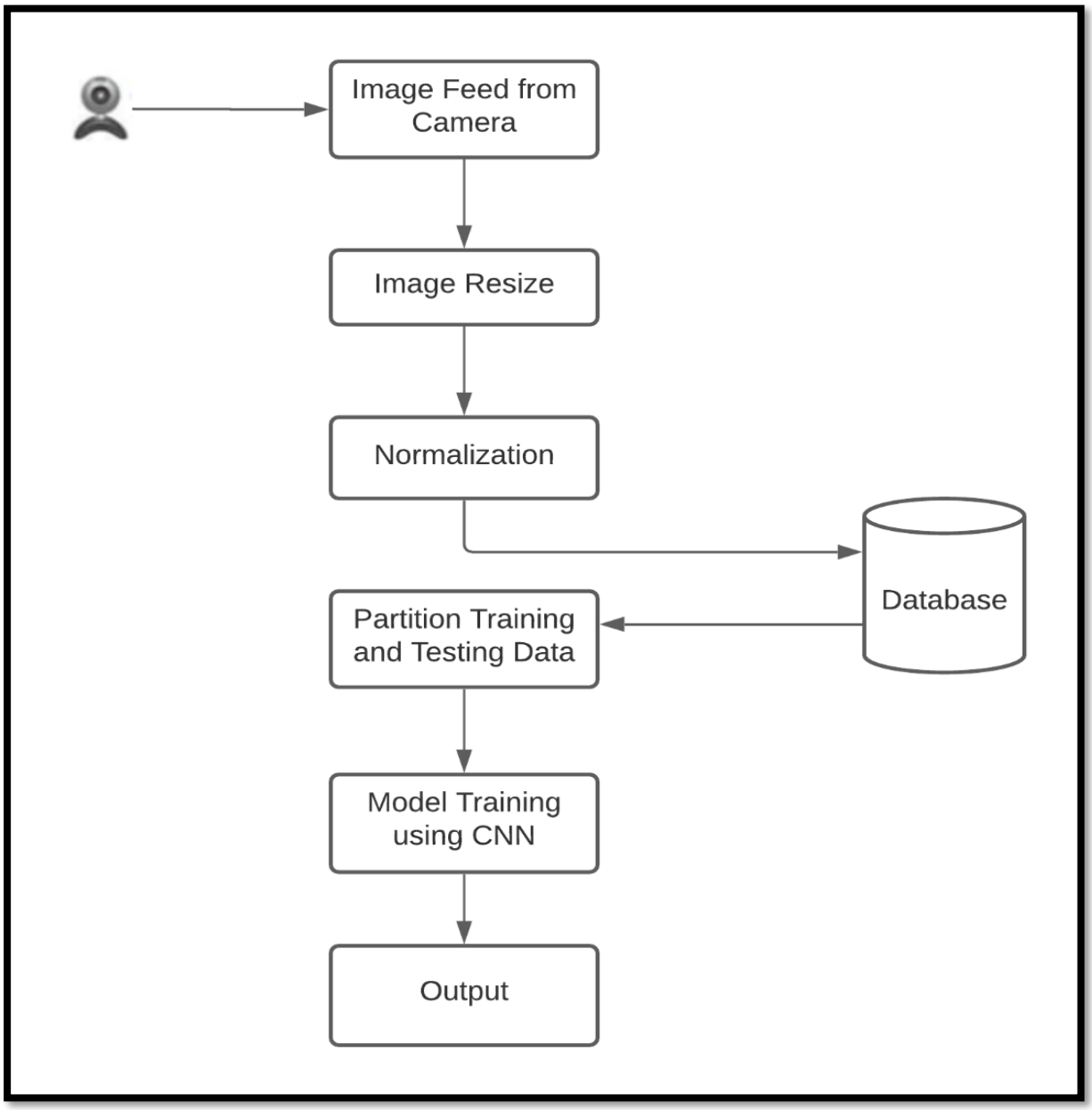
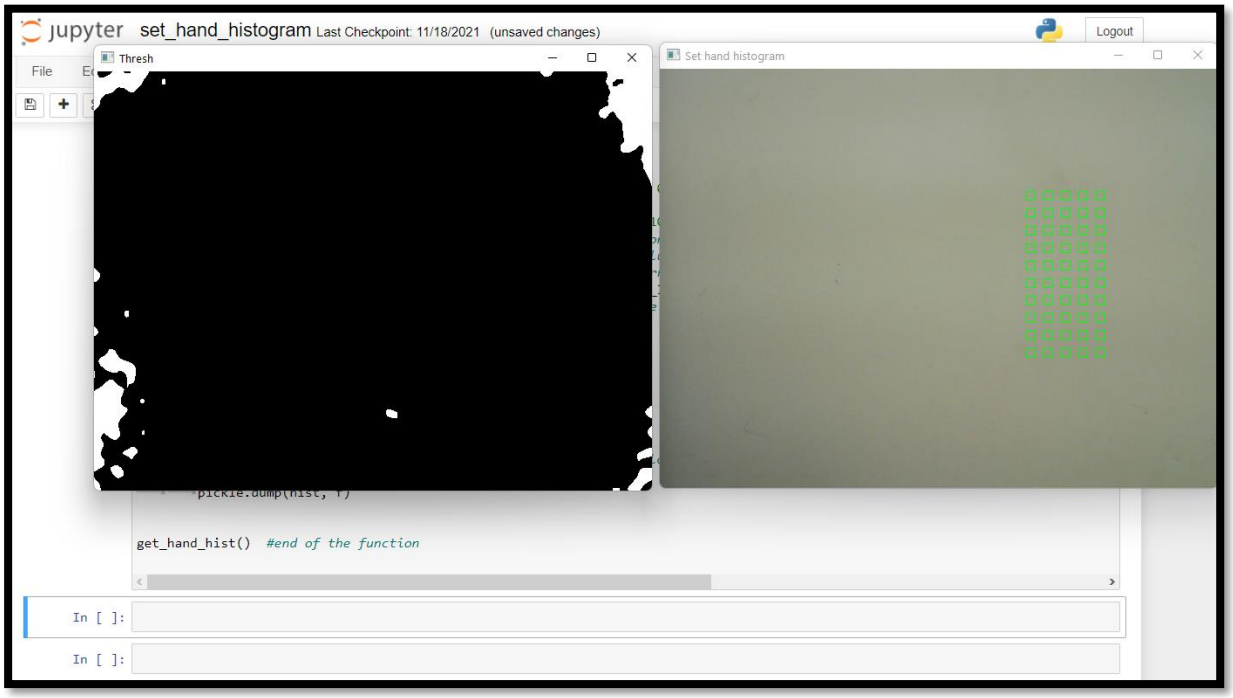


Fig 3.16. Flowchart of Sign Language Recognition

4. PERFORMANCE EVALUATION

4.1 Set hand histogram

In the Set hand Gesture, we have created a window which will take input from the camera feed. We have created 10 vertical and 5 horizontal tables of boxes making it to 50 boxes so as we can connect the neural network to the hand input.

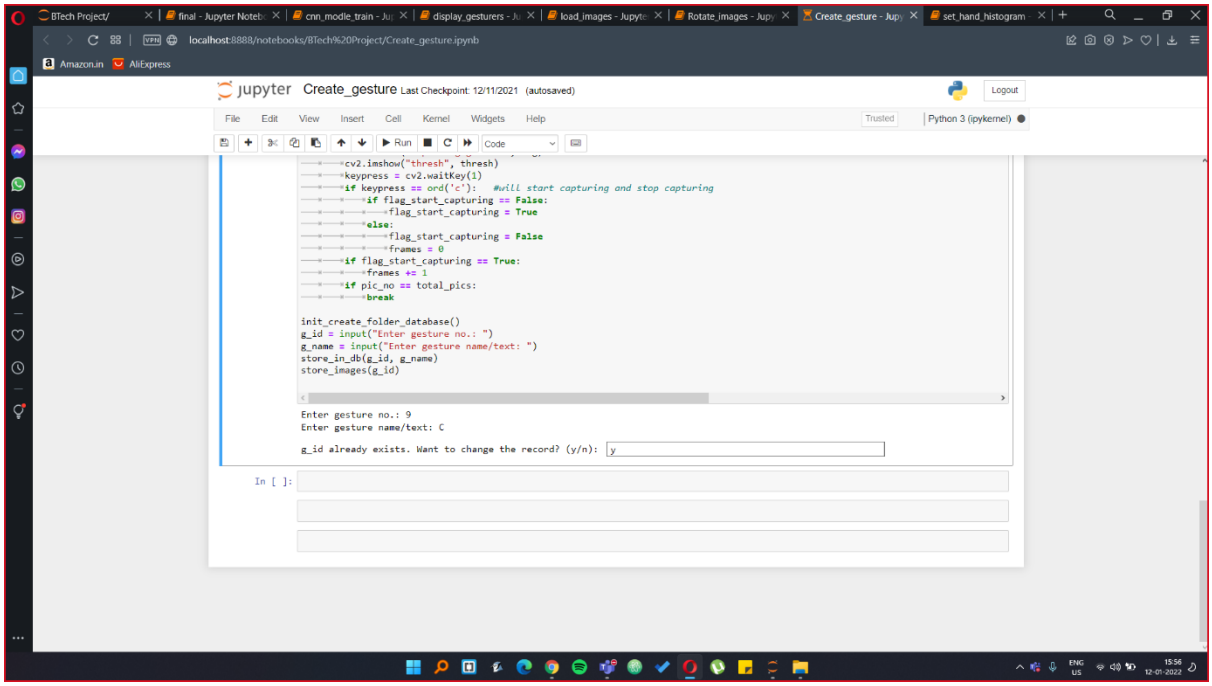


Screenshot 4.1. Set hand histogram

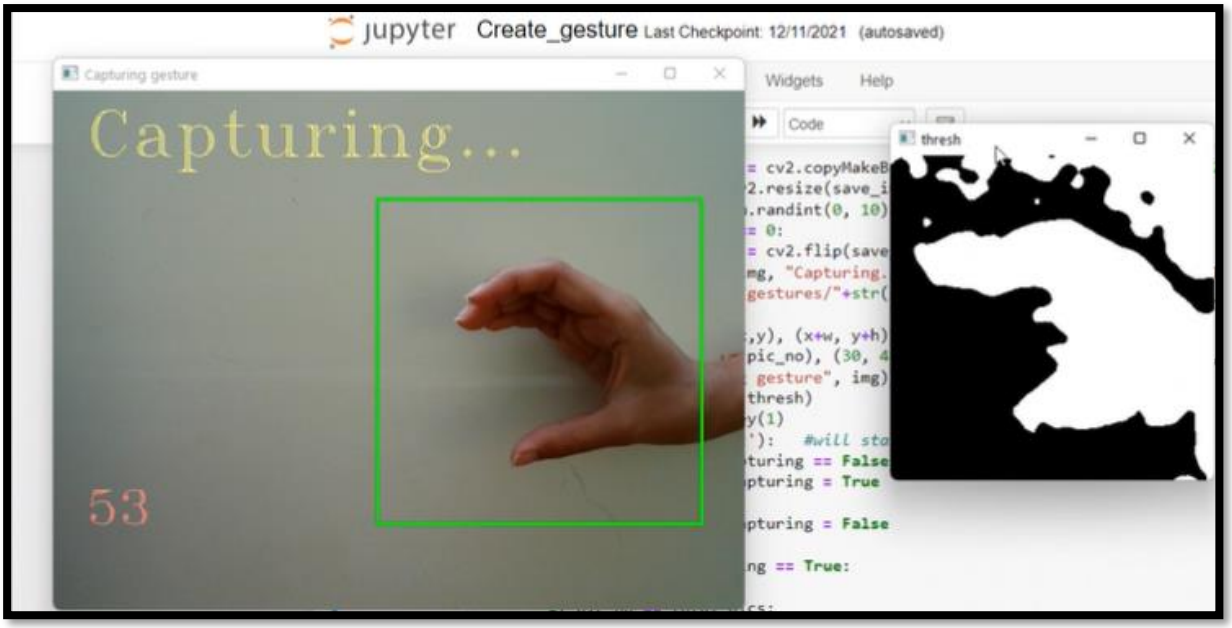
4.2 Create Gesture

In this stage we have created a database which takes the input from the user as g_id that is Gesture id and the g_name that is the gesture name of the input taken from the camera. If the g_id is already represented then it will ask whether to change it or not.

If not present then it will start to take the input as a form is snaps till the count reaches to 1200.



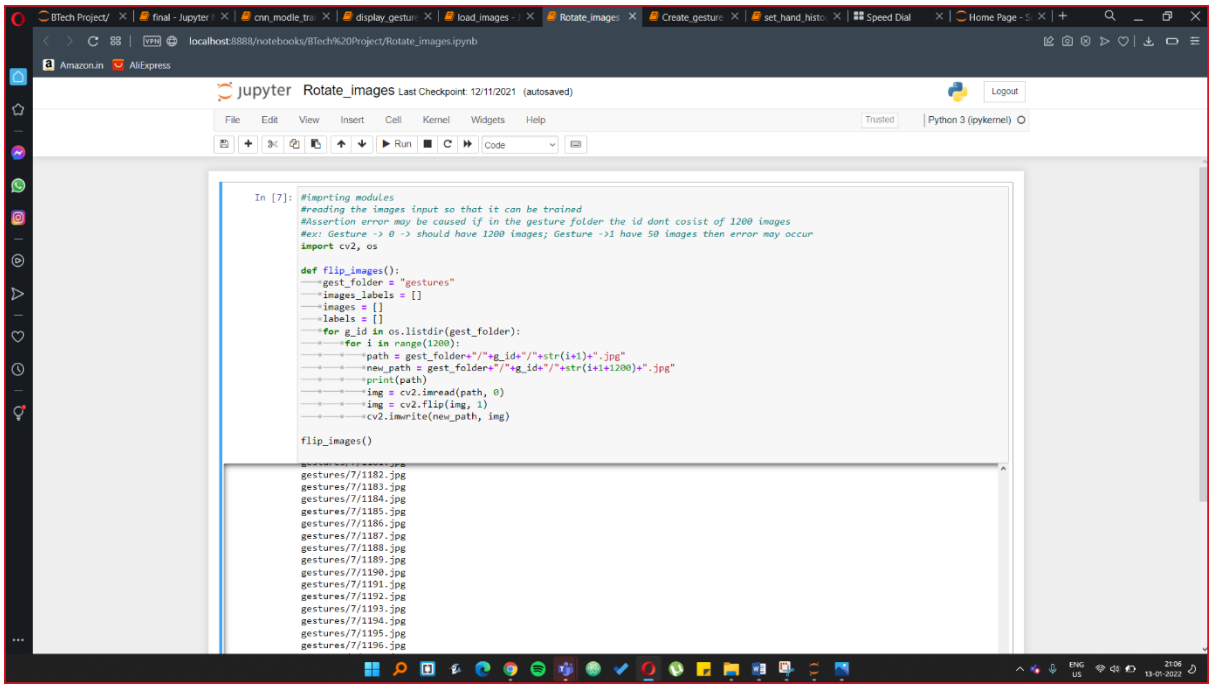
Screenshot 4.2. Creation of database



Screenshot 4.3. Create gestures using OpenCV

4.3 Rotate images

In this we flip the input images which will take the count to 2400 as we cannot specify whether the input is from left hand or right hand so we flip it to avoid the consciousness

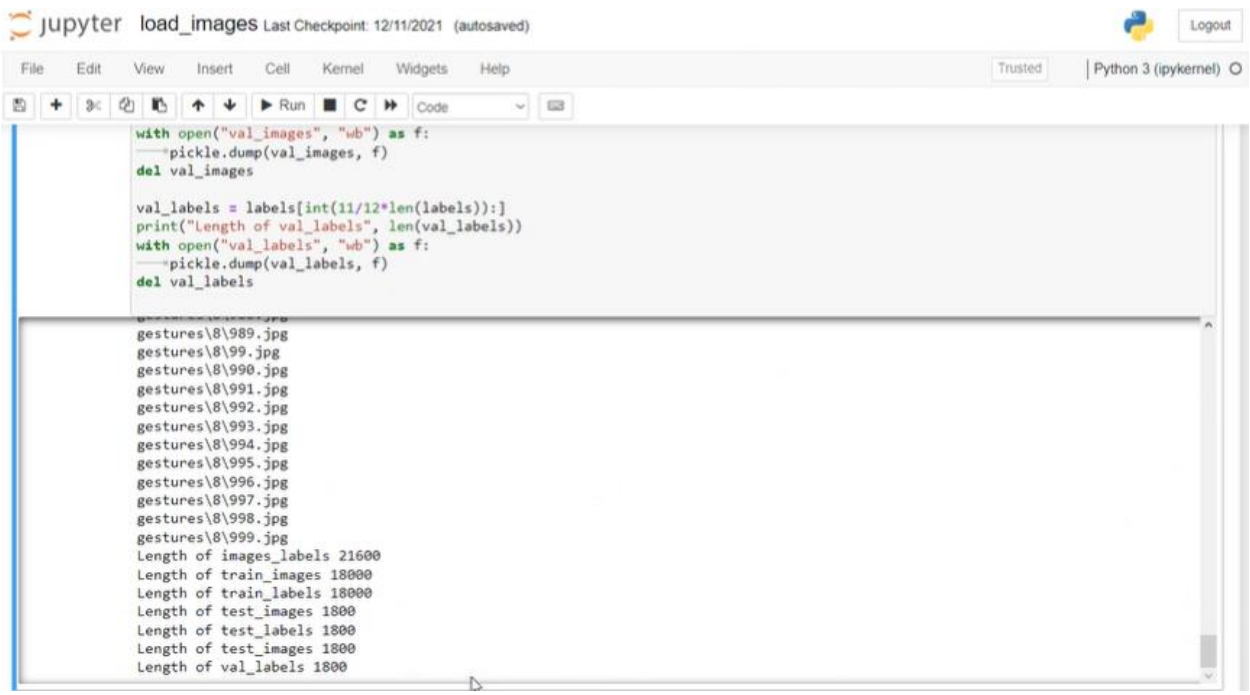


Screenshot 4.4. Rotate image code

4.4 Load Images

In this we divide the images into testing, training labels.

As we need to give training input in order to train the model.



Screenshot 4.5. Load images

4.5 Display Gestures

With the help of display gesture module, we can view all the gestures that has been captured. So that the user can also know the images outcome after testing and training the data.

Screenshot 4.6. Display gestures

4.6 CNN Model Training

In this stage we have taken relu as the activation source for training the cnn model.

We have trained the CNN model using keras and got accuracy of about 99%.

Convolutional Neural Network (CNN):

1. A convolutional neural network, or CNN, is a deep learning neural network sketched for processing structured arrays of data such as portrayals.

2. CNN are very satisfactory at picking up on design in the input image, such as lines, gradients, circles, or even eyes and faces.

3. The strength of a convolutional neural network comes from a particular kind of layer called the convolutional layer.

4. The construction of a convolutional neural network is a multi-layered feed-forward neural network, made by assembling many unseen layers on top of each other in a particular order.

5. It is the sequential design that give permission to CNN to learn hierarchical attributes.

Activation Function:

In a neural network, the activation function is responsible for transforming the summed weighted input from the node into the activation of the node or output for that input.

Relu Activation Function:

The rectified linear activation function or ReLU for short is a piecewise linear function that will output the input directly if it is positive, otherwise, it will output zero. It has become the default activation function for many types of neural networks because a model that uses it is easier to train and often achieves better performance.

Screenshot 4.7. Model Training

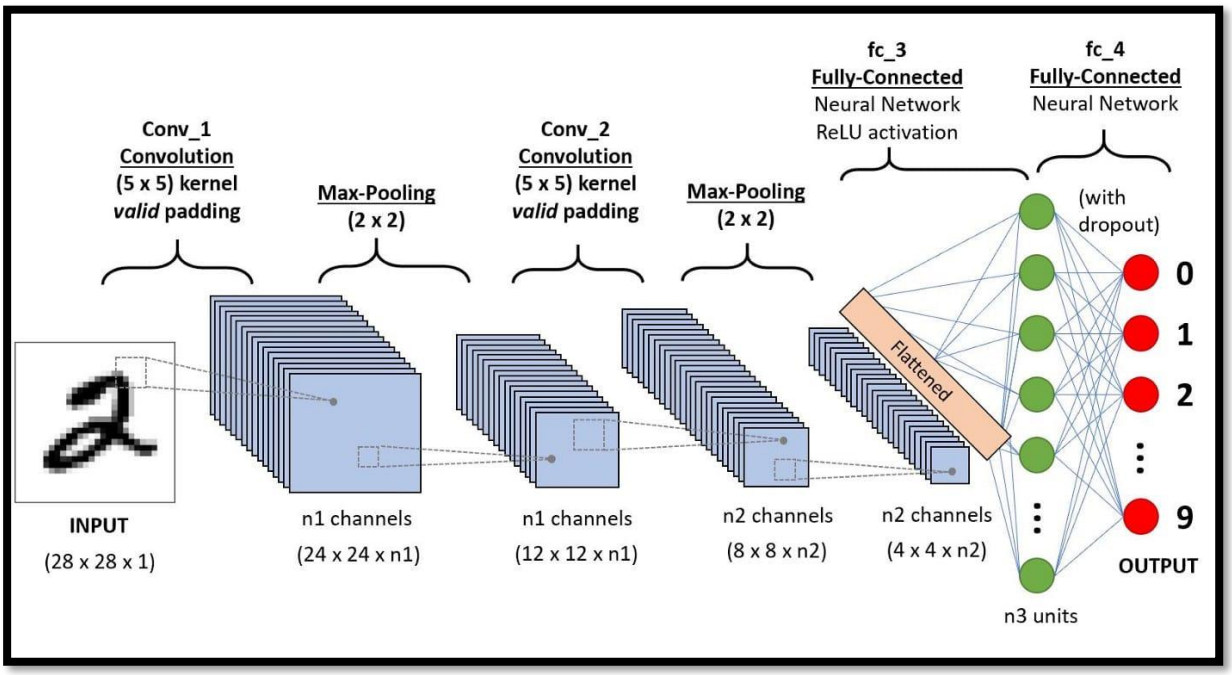
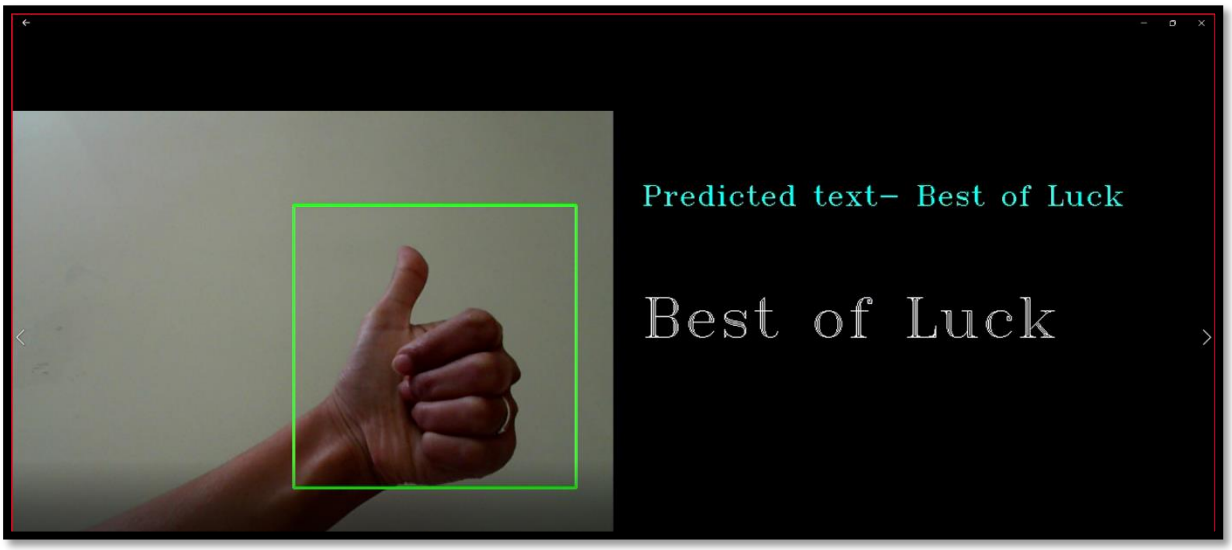


Fig 4.1. Working of CNN

4.7 Final

In the final stage we have club all the code sources in one code so that we can predict the output easily.

When the model code is running it will have a screen which will have a window for camera input and a predicted test label. In this it will try to guess the text label. And if the input satisfy the trained images the output will be displayed on the next line with an audio voice.



Screenshot 4.8. Final Output

5. CONCLUSION

- The aim of this project is to predict the ASL (American Sign Language) alphanumeric hand-gestures in real-time. Our work done above shows that this problem of predicting the hand gestures properly so that it should be understood by the deaf people is appropriate.
- We have taken the gestures through the video from camera and converted them to the grayscale images by reducing the noise of image thus, increasing the clarity so that the prediction is more accurate.
- The model used for training is Convolutional Neural Networks i.e., CNN. CNN takes the images and divides them into training and testing tests thereby generating the epoch.
- At the end of training, when the module is tested, we were able to achieve an accuracy of 99% and the error was 1% since the input gestures captured by us were very clear resulting in maximum accuracy.
- Our model showed good accuracy with around 8 gestures being recorded.

REFERENCES

- https://www.sciencedirect.com/science/article/pii/S1877050917320720 Sited on: 14.10.'21
- https://www.researchgate.net/publication/355915977_SIGN_LANGUAGE_RECOGNITION Sited on: 11.11.'21
- https://www.ijert.org/sign-language-recognition-system-using-convolutional-neural-network-and-computer-vision Sited on: 25.11.'21
- https://www.eecg.utoronto.ca/~jayar/mie324/asl.pdf Sited on 05.12.'21
- https://edu.authorcafe.com/academies/6813/sign-language-recognition Sited on 15.12'21
- https://cse.iitk.ac.in/users/cs365/2015/_submissions/vinsam/report.pdf Sited on 03.01'22

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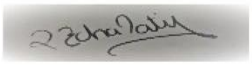
We are extremely thankful to Dr. Ulhas Shiurkar, Director, Deogiri Institute of Engineering, and management Studies Aurangabad, for providing me infrastructural facilities to work in, without which this work would not have been possible.

Signature(s) of Students

Richa Vijay Patil 

Riteeka Amol Purnekar

Nilesh Sharad Mahajan